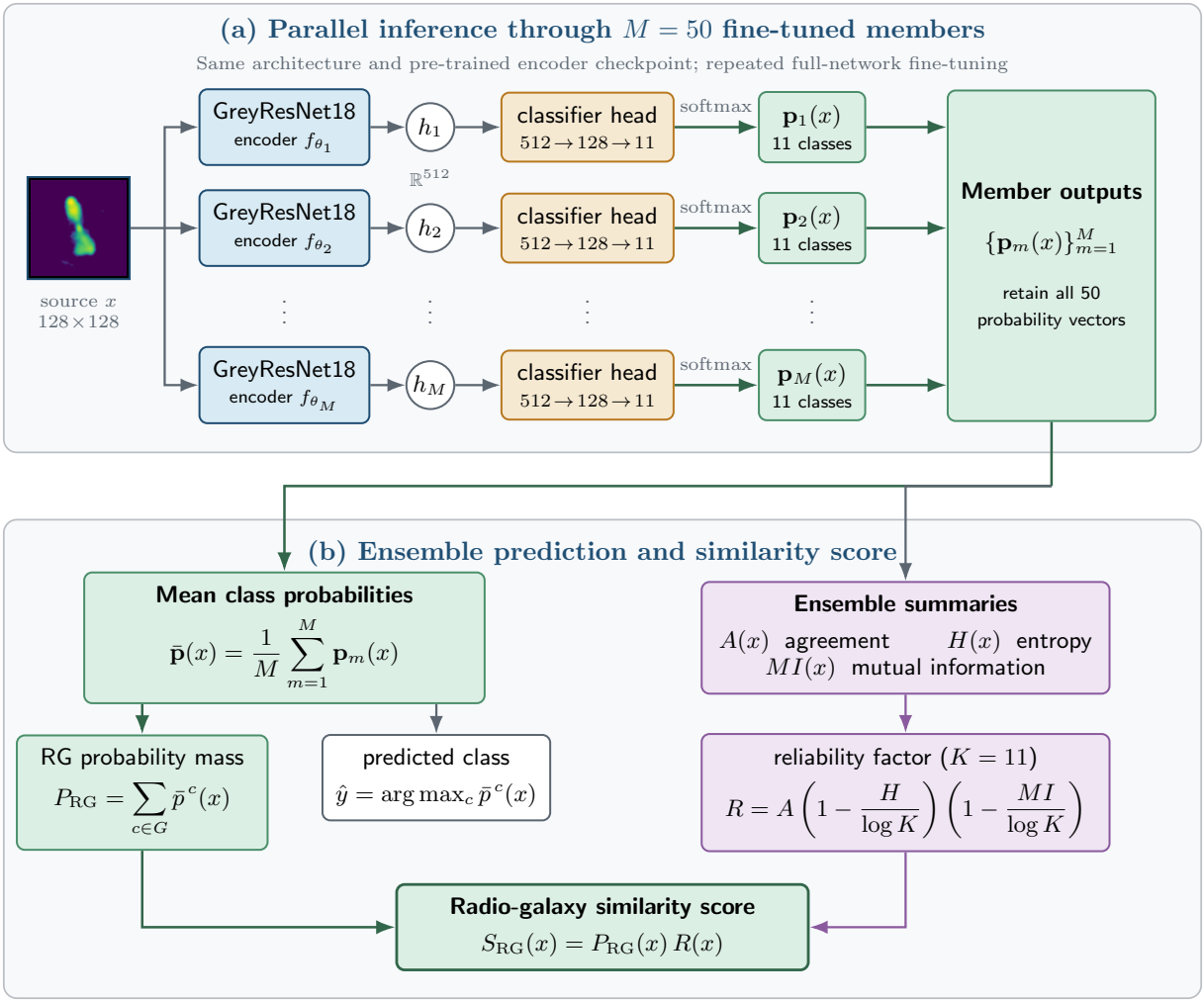

(a) Parallel inference through $M = 50$ fine-tuned members
Same architecture and pre-trained encoder checkpoint; repeated full-network fine-tuning
source x
128×128
GreyResNet18
encoder f_{θ_1}
GreyResNet18
encoder f_{θ_2}
GreyResNet18
encoder f_{θ_M}
h_1
h_2
h_M
$\mathbb{R}^{512}$
classifier head
$512 \to 128 \to 11$
classifier head
$512 \to 128 \to 11$
classifier head
$512 \to 128 \to 11$
softmax
softmax
softmax
$\mathbf{p}_1(x)$
11 classes
$\mathbf{p}_2(x)$
11 classes
$\mathbf{p}_M(x)$
11 classes
Member outputs
$\{\mathbf{p}_m(x)\}_{m=1}^{M}$
retain all 50
probability vectors
(b) Ensemble prediction and similarity score
Mean class probabilities
$\bar{\mathbf{p}}(x) = \frac{1}{M} \sum_{m=1}^{M} \mathbf{p}_m(x)$
RG probability mass
$P_{\mathrm{RG}} = \sum_{c \in G} \bar{p}^{\,c}(x)$
predicted class
$\hat{y} = \arg\max_c \bar{p}^{\,c}(x)$
Ensemble summaries
$A(x)$ agreement $H(x)$ entropy
$MI(x)$ mutual information
reliability factor ($K = 11$)
$R = A \left(1 - \frac{H}{\log K}\right) \left(1 - \frac{MI}{\log K}\right)$
Radio-galaxy similarity score
$S_{\mathrm{RG}}(x) = P_{\mathrm{RG}}(x)\, R(x)$